OPERATING SYSTEMS – ASSIGNMENT 1

TASK 12 – Re-evaluate and Adapt the Design (8 Marks)
CO1–CO5 | Bloom's Level: K5 – Evaluate

Introduction

The hospital has added remote patient-monitoring devices in patients' homes and moved patient records to a cloud service shared with partner hospitals. These changes affect communication, security, availability and the original operating-system plan. The design should be modified rather than completely replaced.

A. Impact analysis (2 Marks)

Operating-system types affected: Home monitoring devices need an embedded real-time OS suitable for local sensor reading and alarm handling. The shared cloud records service needs a reliable server OS and a distributed system connecting the hospitals.
Services affected: Remote devices need secure network communication, device management, error detection and reliable data transfer. Cloud records need authentication, authorisation, storage management, backup, recovery, auditing and communication services.
Protection affected: Data travels across public networks, so encryption, secure authentication, device identity, access control and audit logging become more important. Only authorised users and devices should access patient information.

B. Evaluation of the original design (2 Marks)

The original local design remains suitable for bedside monitors, tablets, workstations and MRI processing. However, it needs additional support for home devices and cloud records. Home internet connections may be slow or unavailable, so remote readings cannot depend on continuous connectivity. A home device should continue its local monitoring functions even if the network fails. Cloud records can make sharing easier across hospitals, but they introduce dependence on internet connectivity, cloud-service availability and secure identity management. The original plan therefore needs modification to address these risks.

C. Adaptation or redesign (2 Marks)

1. Keep the embedded real-time OS on bedside monitors and use a suitable embedded real-time OS on home monitoring devices. Local alarm functions should continue even without internet access.
2. Store readings temporarily on the device when connectivity is unavailable and transmit them securely when the connection returns. The device should identify timestamps and detect duplicate or missing data.
3. Use secure communication, device authentication and encryption when home devices send readings to the hospital service.
4. Move shared patient records to a cloud-based distributed service with controlled access for the main hospital and partner hospitals.
5. Use backups, replication, monitoring and disaster-recovery procedures to improve cloud-service availability.
6. Keep tablets and workstations protected with role-based access, updates and audit logging. MRI processing can remain on a suitable local multiprocessor server or use approved cloud resources when appropriate.

D. Evidence-based defence (2 Marks)

The design should be **modified**, not fully redesigned. The real-time monitor design should be retained because local, predictable alarm response is still essential. The home devices should add secure communication and temporary data storage so that a network interruption does not stop local monitoring. The central record system should be modified to use a cloud-based distributed service because multiple hospitals need shared access. Replication and backups improve availability, while encryption and authentication protect patient information. The trade-off is increased dependence on networks and cloud providers, so local operation, recovery plans and service monitoring are necessary.

Revised design summary

Bedside monitors: Embedded real-time OS; local alarm generation.
Home monitoring devices: Embedded real-time OS; local monitoring, temporary storage and secure transmission.
Nurses' tablets: Mobile OS; secure access to authorised records.
Doctors' workstations: Desktop OS; authenticated clinical applications.
Shared records: Cloud-based distributed service; encryption, access control, replication and backups.
MRI imaging: Multiprocessor system with high-throughput I/O; local or approved cloud processing depending on workload and policy.
Hospital connectivity: Secure network links with monitoring and recovery procedures.

Conclusion

The best decision is to modify the original design. Retain local real-time monitoring and specialised systems, add secure embedded home-monitoring devices, and use a protected cloud-based distributed records service. This supports remote care and record sharing while protecting patient safety and confidentiality.

Prepared from the corresponding task in the uploaded CSA04 Operating Systems Assignment 1 document. Add your name, register number, required screenshots and your own measured evidence wherever requested before submission.